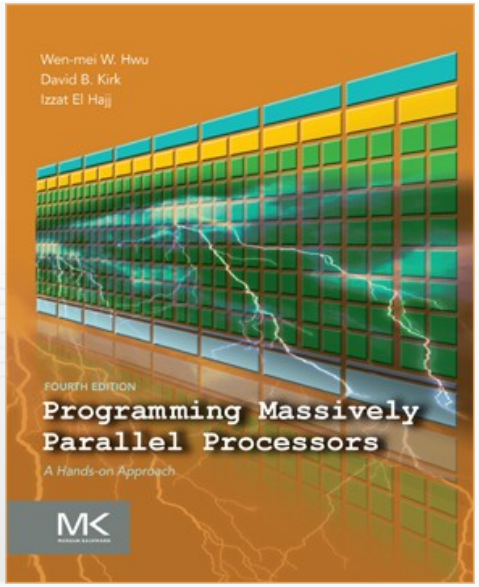
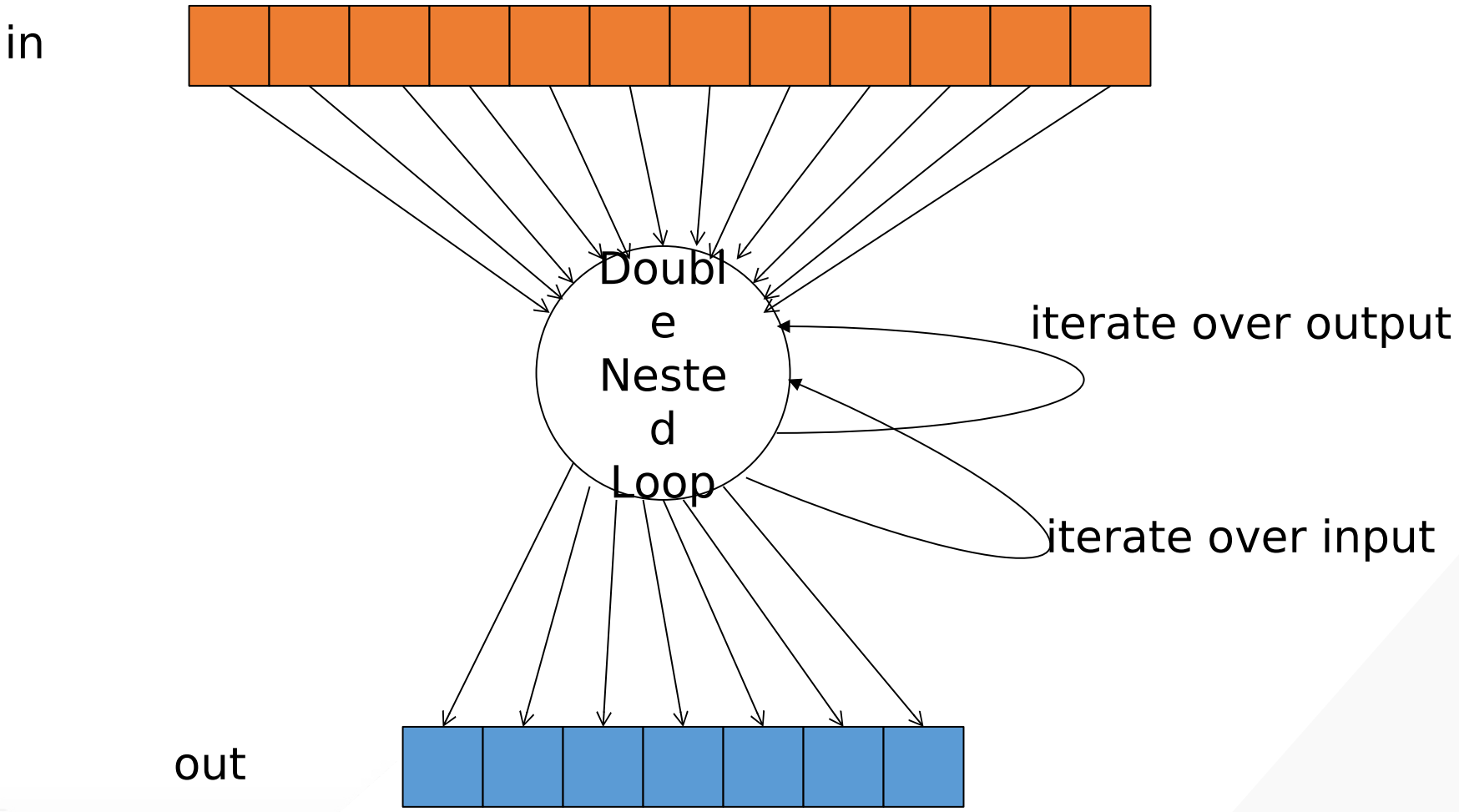
Programming Massively Parallel Processors

A Hands-on Approach

CHAPTER 18 › Application Case Study – Electrostatic Potential Map

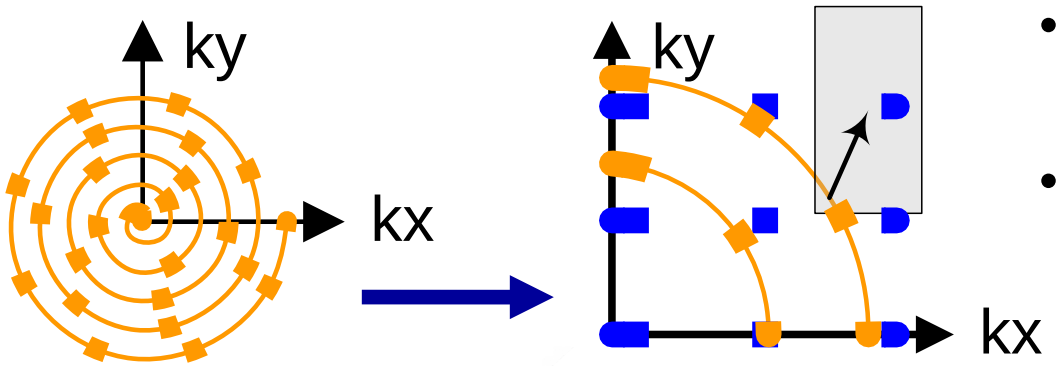


A Common Sequential Computation Pattern



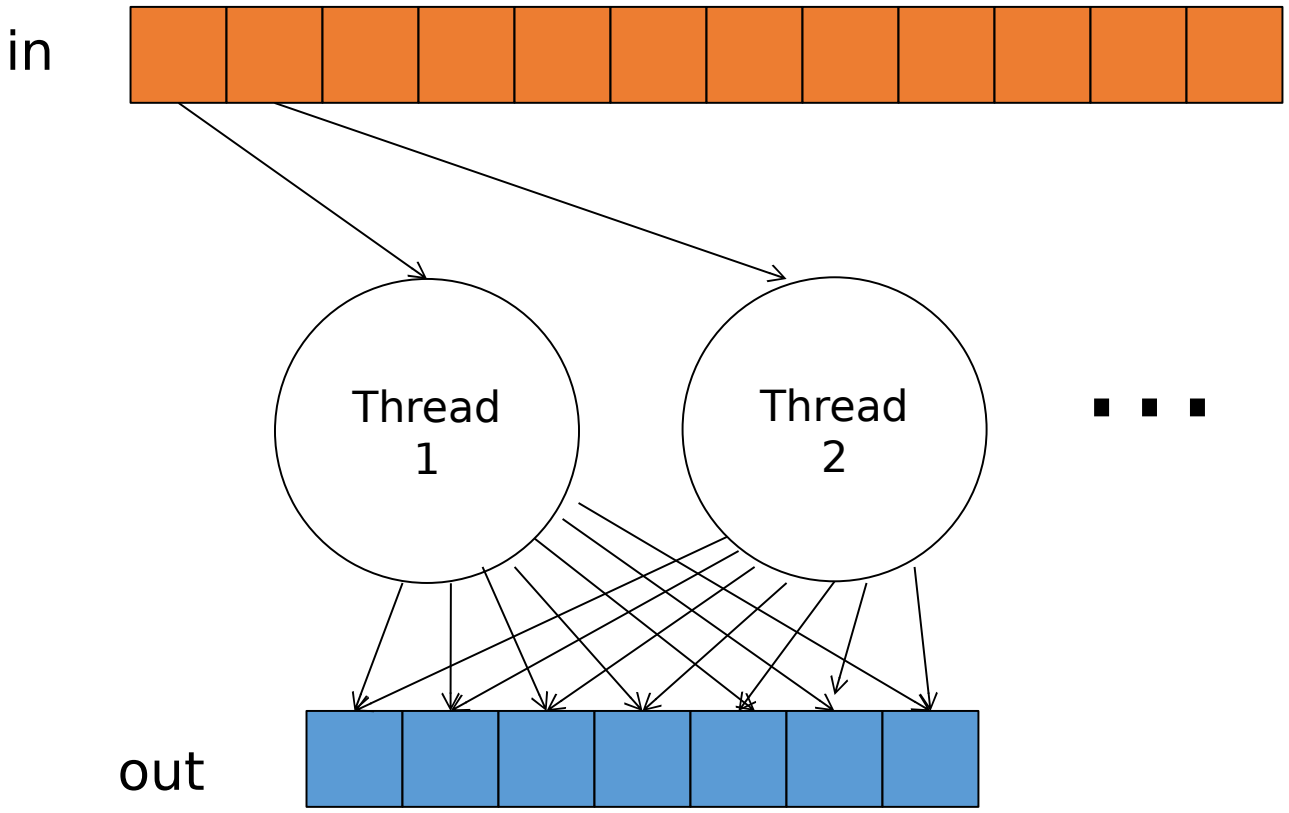
A Simple Code Example

```
for (m = 0; m < M; m++) {

  for (n = 0; n < N; n++) {

    out[n] += f(in[m], m, n);
  }
}
```



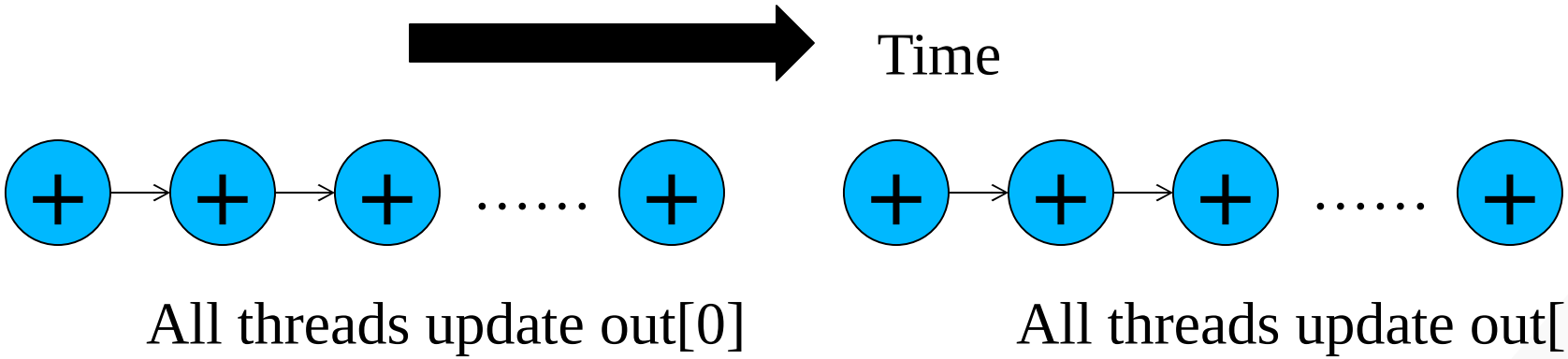
- Input data “in”
  - M = # scan points
- Output data “out”
  - N = # regularized scan points
- Complexity is O(MN)
- Output tends to be more regular than input

Scatter Parallelization

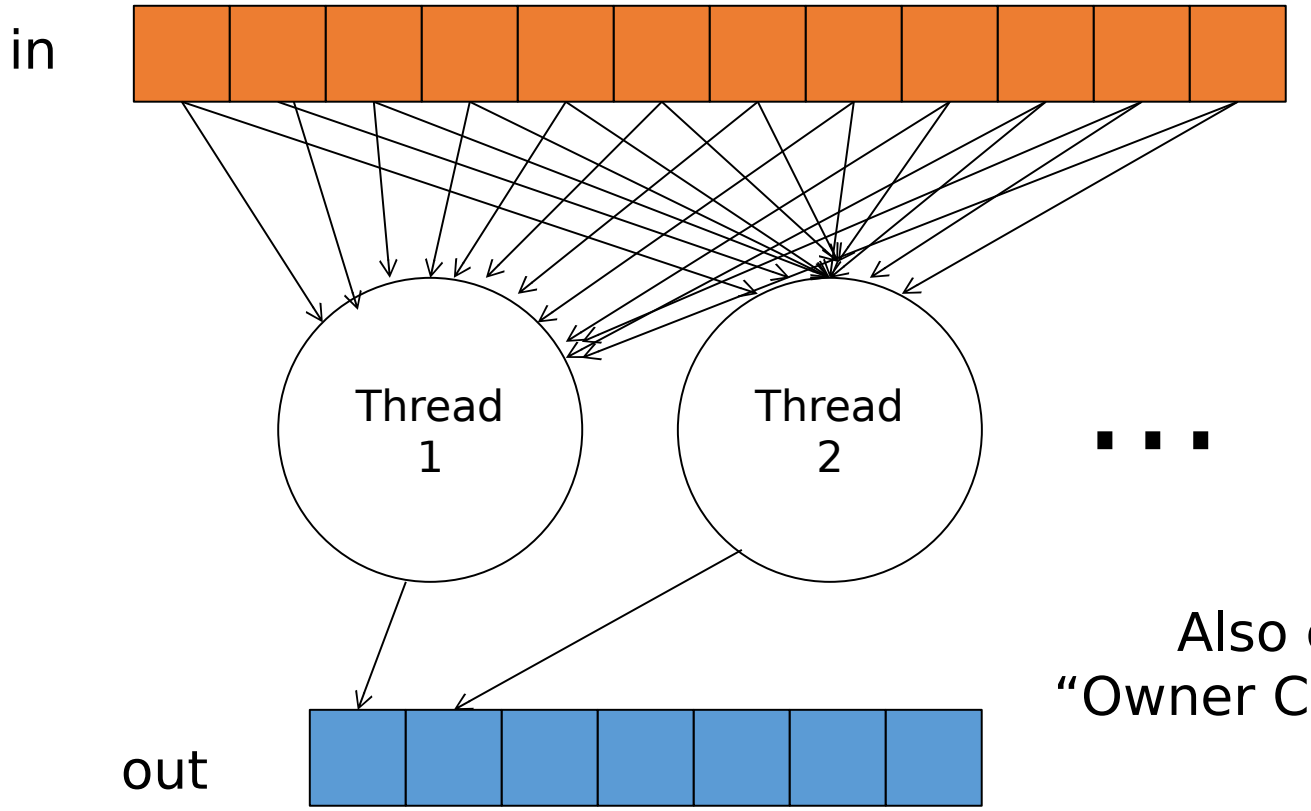


Scatter can be very slow.

- All threads have conflicting updates to the same out elements
  - Serialized with atomic operations
  - Potentially very costly (slow) for large number of threads



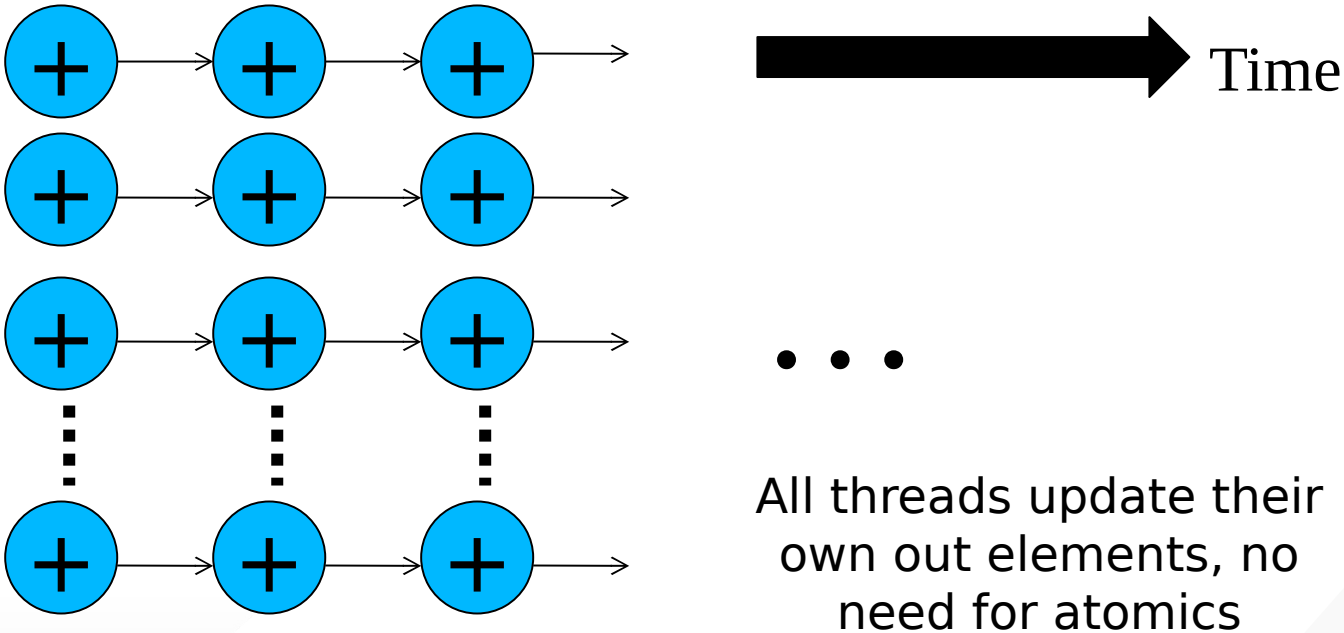
Gather Parallelization



Also called the
“Owner Computes” rule

Gather can be very fast.

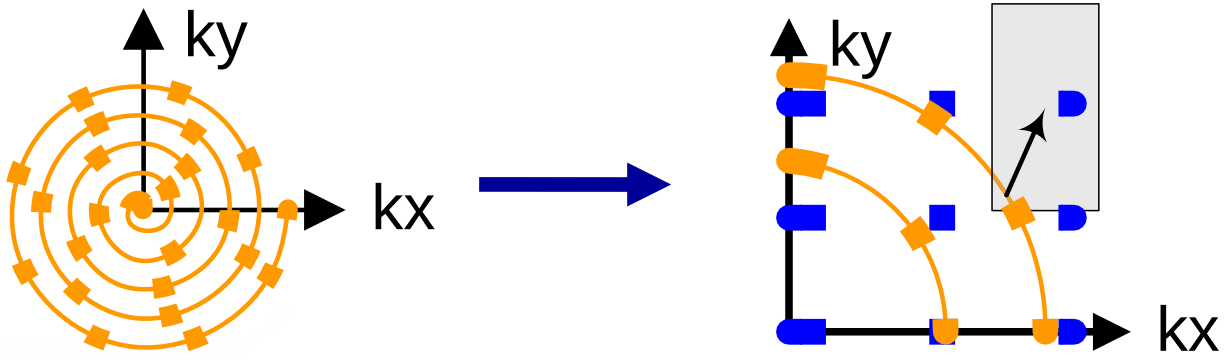
- All threads can read the same “in” elements
  - No serialization
  - Can even be efficiently consolidated through caches or scratchpad (shared) memories



All threads update their own out elements, no need for atomics

Why is scatter parallelization often used rather

- In practice, each in element does not affect all “out” elements
- Output tends to be much more regular than input
- It is easy to calculate all out elements affected by an in element
  - Harder to calculate all in elements needed for an out element
  - Easy thread kernel code if written in scatter



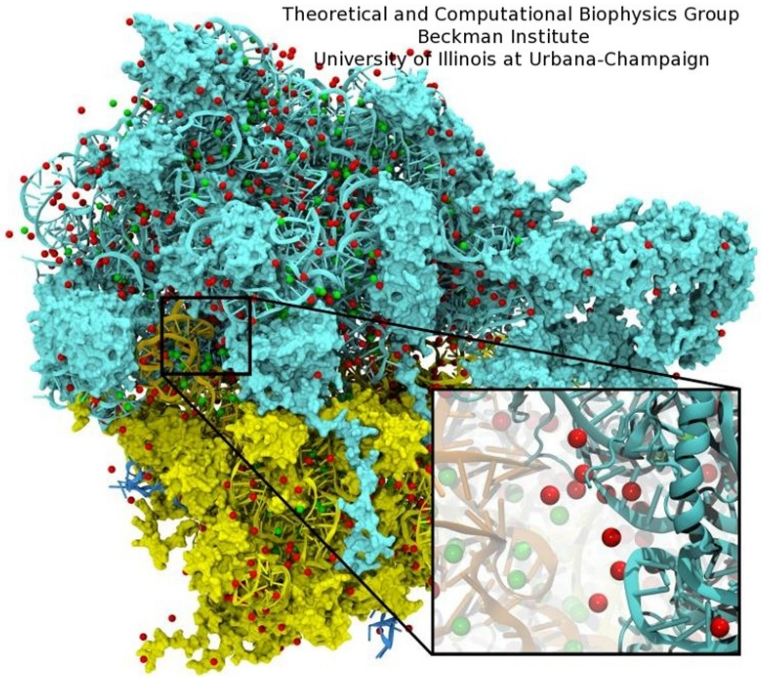
Challenges in Gather Parallelization

- Regularize input elements so that it is easier to find all in elements that affect an out element
  - Cut-off Binning Lecture

- For this lecture, we assume that all in elements affect all out elements

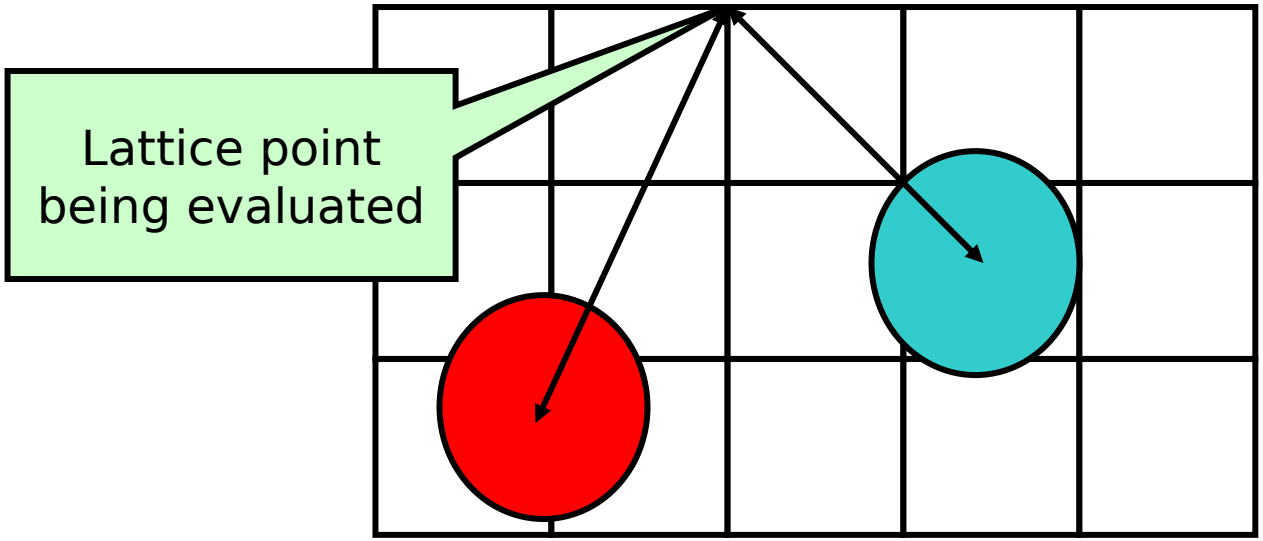
Example – Ion Placement in Molecular Modeling

- Biomolecular simulations attempt to replicate *in vivo* conditions *in silico*
- Model structures are initially constructed in vacuum
- Solvent (water) and ions are added as necessary to reproduce the required biological



Ion Placement Process (Step 1)

- Calculate an initial electrostatic potential map around the simulated structure considering the contributions of all atoms
  - Very time consuming in VMD, focus of our example.



Ion Placement Process (Step 2)

- Ions are then placed one at a time:
  - Find the point with the minimum potential value
  - Add a new ion atom at location of minimum potential
  - Add the potential contribution of the newly placed ion to the entire map
  - Repeat until the required number of ions have been added

Overview of Direct Coulomb Summation (DCS) Algorithm

- Most accurate way to compute the electrostatic potentials on a grid, ideally suited for the GPU
  - All atoms affect all map lattice points

- For each lattice point, sum potential contributions for all atoms in the simulated structure:

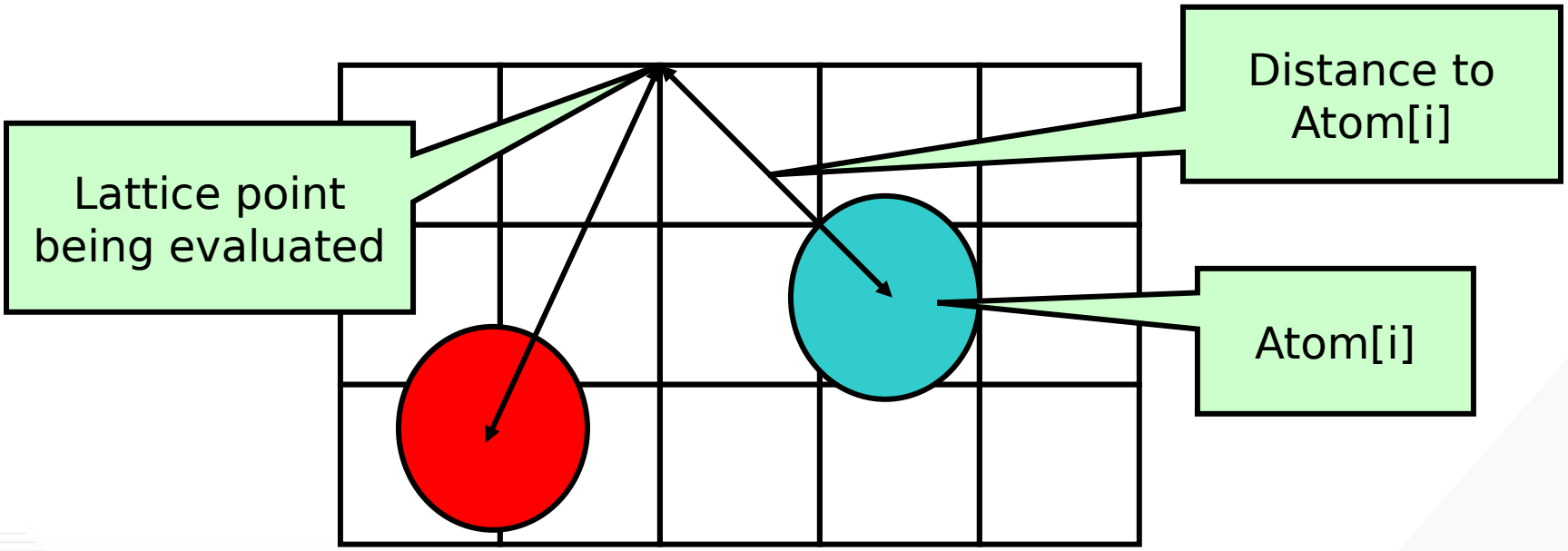
  potential += charge[i] / (distance to atom[i])

- Approximation-based methods such as cut-off summation can achieve much higher performance at the cost of some numerical accuracy and flexibility
  - Will cover these later in the Binning Lecture

Direct Coulomb Summation (DCS) Algorithm Details

- At each lattice point, sum potential contributions for all atoms in the simulated structure:

  potential += charge[i] / (distance to atom[i])



Electrostatic Potential Map Calculation Function Overview

- Each call calculates an x-y slice of the energy map
  - *energygrid* – pointer to the entire potential map
  - *grid – the x, y, z dimensions of the potential map*
  - *gridspacing – modeled physical distance between grid points*
  - *z – the z coordinate of the x-y slice being calculated*
  - *atoms – array of x, y, z coordinates and charge of atoms*
  - *numatoms – number of atoms in atoms array*

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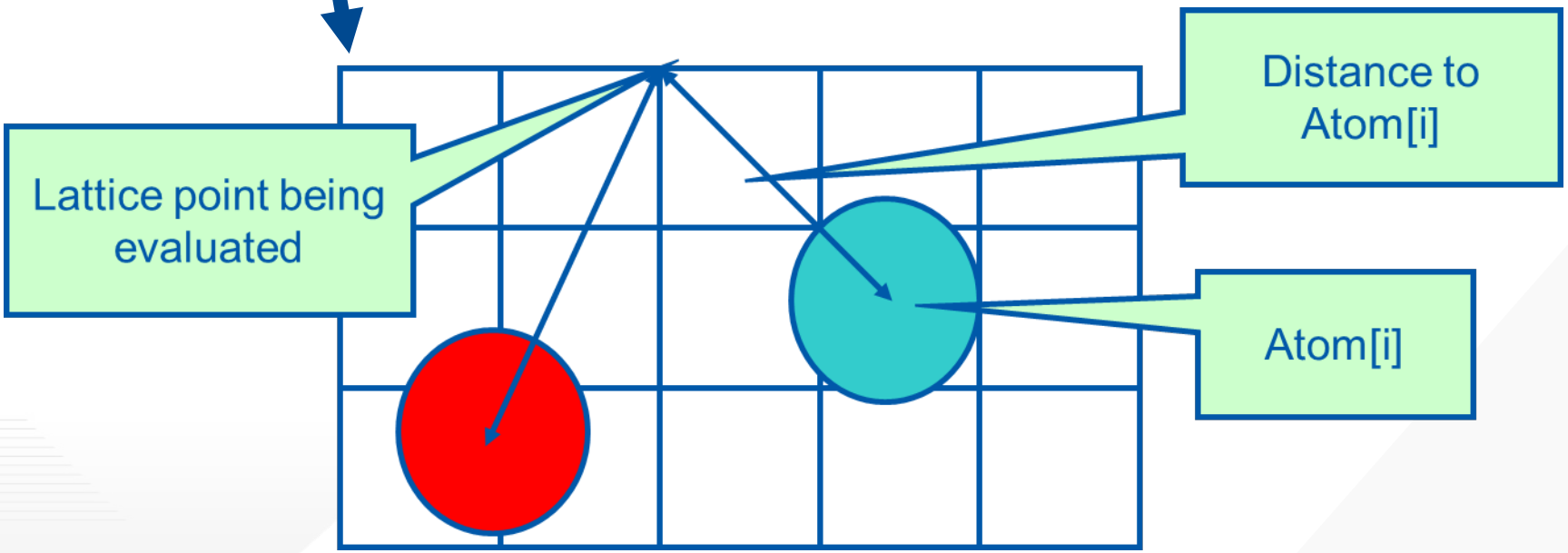

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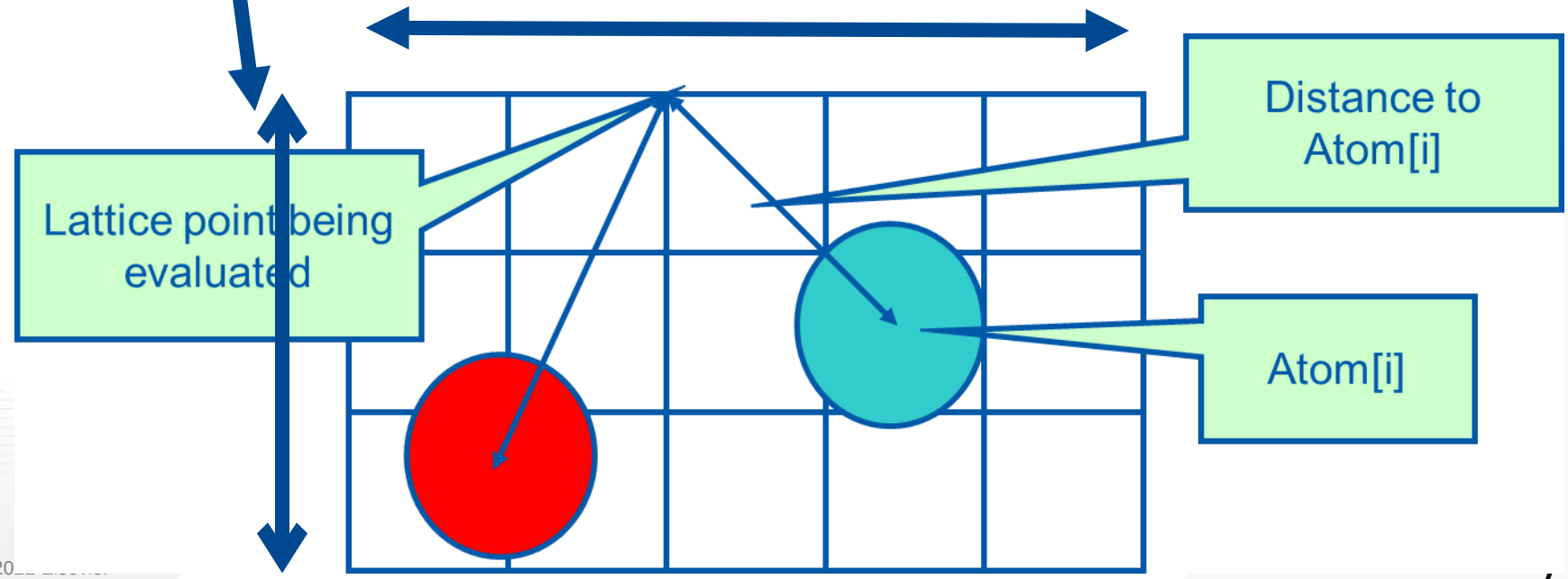

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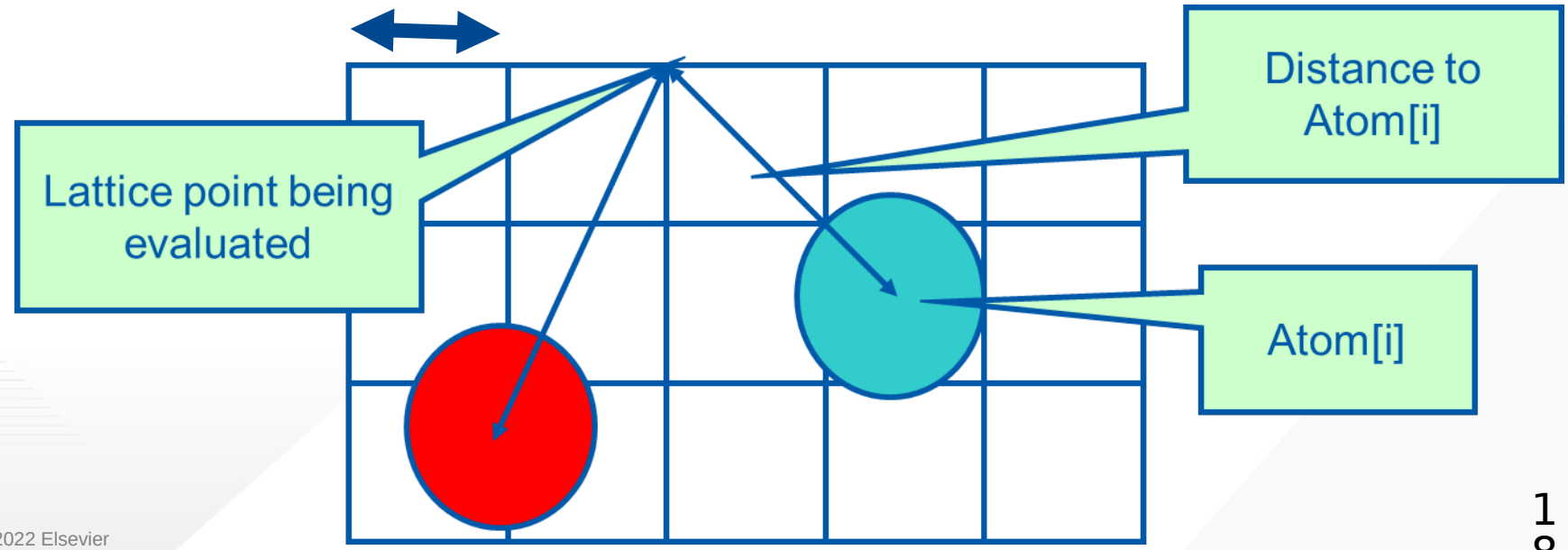

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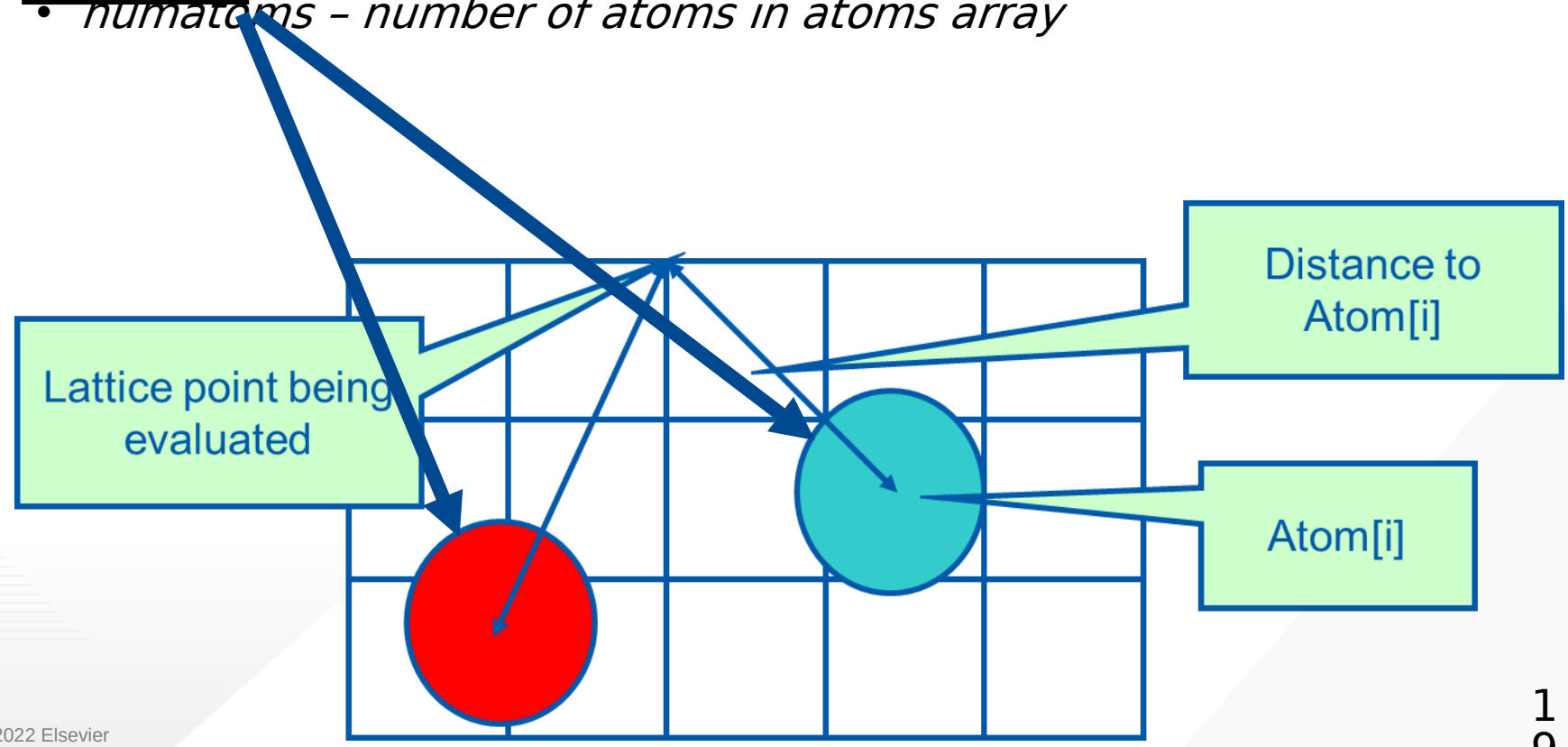

An Intuitive Sequential C Version

```
void cenergy(float *energygrid, dim3 grid, float gridspacing, float z, const float
*atoms, int numatoms) {
  int atomarrdim = numatoms * 4; //x,y,z, and charge info for each atom
  for (int n=0; n<atomarrdim; n+=4){ //calculate potential contribution of each atom
    float dz = z - atoms[n+2];  // all grid points in a slice have the same z value
    float dz2 = dz*dz;
    int grid_slice_offset = (grid.x*grid.y*z) / gridspacing;
    float charge = atoms[n+3];
    for (int j=0; j<grid.y; j++) {
      float y = gridspacing * (float) j;
      float dy = y - atoms[n+1];  // all grid points in a row have the same y value
      float dy2 = dy*dy;
      int grid_row_offset =  grid_slice_offset+ grid.x*j;
      for (int i=0; i<grid.x; i++) {
         float x = gridspacing * (float) i;
         float dx = x - atoms[n    ];
         energygrid[grid_row_offset + i] += charge / sqrtf(dx*dx + dy2+ dz2);
      }
    }
  } }
```

Outer loop iterates over all atoms input oriented

An Intuitive Sequential C Version (cont.)

```
void cenergy(float *energygrid, dim3 grid, float gridspacing, float z, const float
*atoms, int numatoms) {
  int atomarrdim = numatoms * 4; //x,y,z, and charge info for each atom
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      float dy2 = dy*dy;
      int grid_row_offset =  grid_slice_offset+ grid.x*j;
      for (int i=0; i<grid.x; i++) {
         float x = gridspacing * (float) i;
         float dx = x - atoms[n    ];
         energygrid[grid_row_offset + i] += charge / sqrtf(dx*dx + dy2+ dz2);
      }
    }
```

For each atom pre-calculate quantities that are invariant across all grid points in the slice

An Intuitive Sequential C Version (cont.)

```
void cenergy(float *energygrid, dim3 grid, float gridspacing, float z, const float
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  int atomarrdim = numatoms * 4; //x,y,z, and charge info for each atom
  for (int n=0; n<atomarrdim; n+=4){ //calculate potential contribution of each atom
    float dz = z -atoms[n+2]; //all grid points in a slice have the same z coordinate
    float dz2 = dz*dz;
    int grid_slice_offset = (grid.x*grid.y*z) / gridspacing;
    float charge = atoms[n+3];
    for (int j=0; j<grid.y; j++) {
      float y = gridspacing * (float) j;
      float dy = y - atoms[n+1];  // all grid points in a row have the same y value
      float dy2 = dy*dy;
      int grid_row_offset =  grid_slice_offset + grid.x * j;
      for (int i=0; i<grid.x; i++) {
         float x = gridspacing * (float) i;
         float dx = x - atoms[n];
         energygrid[grid_row_offset + i] += charge / sqrtf(dx*dx + dy2+ dz2);
      }
    }
  }
```

For each row in the x direction precalculate the quantities that do not vary across all grid points in the row

An Intuitive Sequential C Version (cont.)

```
void cenergy(float *energygrid, dim3 grid, float gridspacing, float z, const float
*atoms, int numatoms) {
  int atomarrdim = numatoms * 4; //x,y,z, and charge info for each atom
  for (int n=0; n<atomarrdim; n+=4){ //calculate potential contribution of each atom
    float dz = z - atoms[n+2];  // all grid points in a slice have the same z  value
    float dz2 = dz*dz;
    int grid_slice_offset = (grid.x*grid.y*z) / gridspacing;
    float charge = atoms[n+3];
    for (int j=0; j<grid.y; j++) {
      float y = gridspacing * (float) j;
      float dy = y - atoms[n+1];  // all grid points in a row have the same y value
      float dy2 = dy*dy;
      int grid_row_offset =  grid_slice_offset+ grid.x*j;
      for (int i=0; i<grid.x; i++) {
         float x = gridspacing * (float) i;
         float dx = x - atoms[n    ];
         energygrid[grid_row_offset + i] += charge / sqrtf(dx*dx + dy2+ dz2);
      }
    }
```

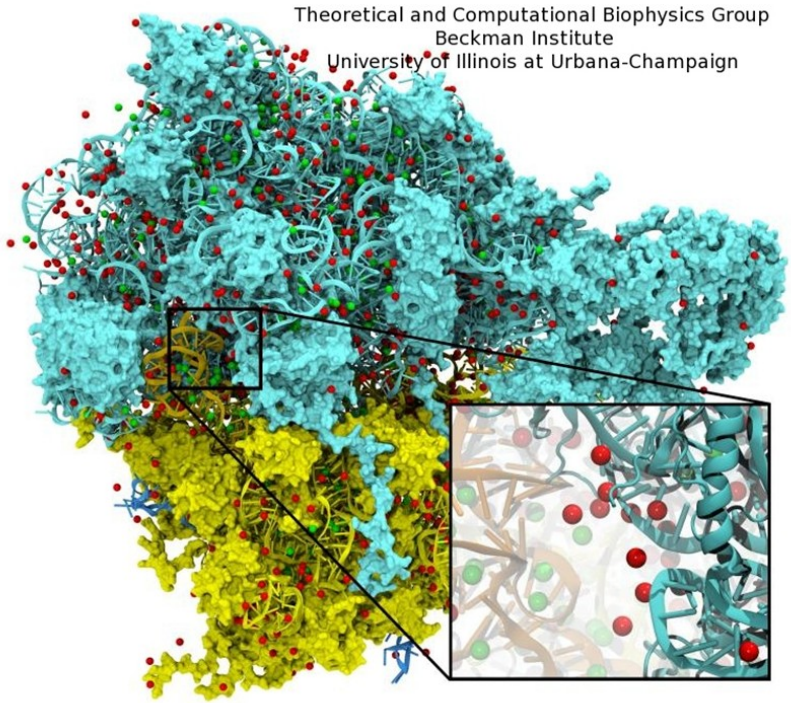
For each grid point in the current row calculate the contribution of potential from the current atom

Summary of Sequential C Version

- Algorithm is input oriented
  - For each input atom, calculate its contribution to all grid points in an x-y slice
- Output (energygrid) is very regular
  - Simple linear mapping between grid point indices and modeled physical coordinates
- Input (atom) is irregular
  - Modeled x,y,z coordinate of each atom needs to be stored in the atom array
- The algorithm is efficient in performing minimal calculations on distances, coordinates, etc.

Irregular Input vs. Regular Output

- Atoms come from modeled molecular structures, solvent (water) and ions
  - Irregular by necessity
- Energy grid models the electrostatic potential value at regularly spaced points
  - Regular by design



CUDA DCS Implementation Overview

- Allocate and initialize potential map memory on host CPU
- Allocate potential map slice buffer on GPU
- Preprocess atom coordinates and charges
- Loop over potential map slices:
  - Copy potential map slice from host to GPU
  - Loop over groups of atoms:
    - Copy atom data to GPU
    - Run CUDA Kernel on atoms and potential map slice on GPU
  - Copy potential map slice from GPU to host
- Free resources

Straightforward CUDA Parallelization

- Use each thread to compute the contribution of an atom to all grid points in the current slice
  - Scatter parallelization

- Kernel code largely corresponds to CPU version with outer loop stripped
  - Each thread corresponds to an outer loop iteration of CPU version
  - numatoms used in kernel launch configuration host code

A Very Slow DCS Scatter Kernel!

```
void  __global__ cenergy(float *energygrid, float *atoms, dim3 grid, float gridspacing, float z) {
    int n = (blockIdx.x * blockDim .x + threadIdx.x) * 4;
    float dz = z - atoms[n+2];  // all grid points in a slice have the same z  value
    float dz2 = dz*dz;
    int grid_slice_offset = (grid.x*grid.y*z) / gridspacing;
    float charge = atoms[n+3];
    for (int j=0; j<grid.y; j++) {
      float y = gridspacing * (float) j;
      float dy = y - atoms[n+1];  // all grid points in a row have the same y value
      float dy2 = dy*dy;
      int grid_row_offset =  grid_slice_offset+ grid.x*j;
      for (int i=0; i<grid.x; i++) {
         float x = gridspacing * (float) i;
         float dx = x - atoms[n    ];
         energygrid[grid_row_offset + i]  += charge / sqrtf(dx*dx + dy2+ dz2));
      }
    }
  }
}
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    float charge = atoms[n+3];
    for (int j=0; j<grid.y; j++) {
      float y = gridspacing * (float) j;
      float dy = y - atoms[n+1];  // all grid points in a row have the same y value
      float dy2 = dy*dy;
      int grid_row_offset =  grid_slice_offset+ grid.x*j;
      for (int i=0; i<grid.x; i++) {
         float x = gridspacing * (float) i;
         float dx = x - atoms[n    ];
         energygrid[grid_row_offset + i]  += charge / sqrtf(dx*dx + dy2+ dz2));
      }
    }
  }
}
```

Needs to be done as an atomic operation

Atomic Operations in CUDA

- Function calls that are translated into single instructions (a.k.a. intrinsics)
  - Atomic add, sub, inc, dec, min, max, exch (exchange), CAS (compare and swap)
  - Read CUDA C programming Guide 4.0 for details

- Atomic Add

  int atomicAdd(int **address**, int **val**);*

  reads the 32-bit word **old** pointed to by **address** in global or shared memory, computes **(old + val)**, and stores the result back to memory at the same address. The function returns **old**.

More Atomic Adds in CUDA

- Unsigned 32-bit integer atomic add
  - unsigned int atomicAdd(unsigned int* address, unsigned int val);
- Unsigned 64-bit integer atomic add
  - unsigned long long int atomicAdd(unsigned long long int* address, unsigned long long int val);
- Single-precision floating-point atomic add (capability > 2.0)
  - float atomicAdd(float* address, float val);

Pros and Cons of the Scatter Kernel

- Pros
  - Follows closely the simple CPU version
  - Good for software engineering and code maintenance
  - Preserves computation efficiency (coordinates, distances, offsets) of sequential code

- Cons
  - The atomic add serializes the execution, very slow!

A Slower Sequential C Version

```
void cenergy(float *energygrid, dim3 grid, float gridspacing, float z, const float *atoms, int numatoms) {

  int atomarrdim = numatoms * 4;
  int k = z / gridspacing;
  for (int j=0; j<grid.y; j++) {
    float y = gridspacing * (float) j;
    for (int i=0; i<grid.x; i++) {
      float x = gridspacing * (float) i;
      float energy = 0.0f;
      for (int n=0; n<atomarrdim; n+=4){//calculate potential contribution of each atom
        float dx = x - atoms[n    ];
        float dy = y - atoms[n+1];
        float dz = z - atoms[n+2];
        energy += atoms[n+3] / sqrtf(dx*dx + dy*dy + dz*dz);
      }
      energygrid[grid.x*grid.y*k + grid.x*j + i] += energy;
    }
  }
}
```

output oriented

Outer loops iterate over all output grid points
Pre-calculate x and y coordinate of the grid point

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```
void cenergy(float *energygrid, dim3 grid, float gridspacing, float z, const float *atoms, int numatoms) {

  int atomarrdim = numatoms * 4;
  int k = z / gridspacing;
  for (int j=0; j<grid.y; j++) {
    float y = gridspacing * (float) j;
    for (int i=0; i<grid.x; i++) {
      float x = gridspacing * (float) i;
      float energy = 0.0f;
      for (int n=0; n<atomarrdim; n+=4){//calculate potential contribution of each atom
        float dx = x - atoms[n    ];
        float dy = y - atoms[n+1];
        float dz = z - atoms[n+2];
        energy += atoms[n+3] / sqrtf(dx*dx + dy*dy + dz*dz);
      }
      energygrid[grid.x*grid.y*k + grid.x*j + i] += energy;
    }
  }
}
```

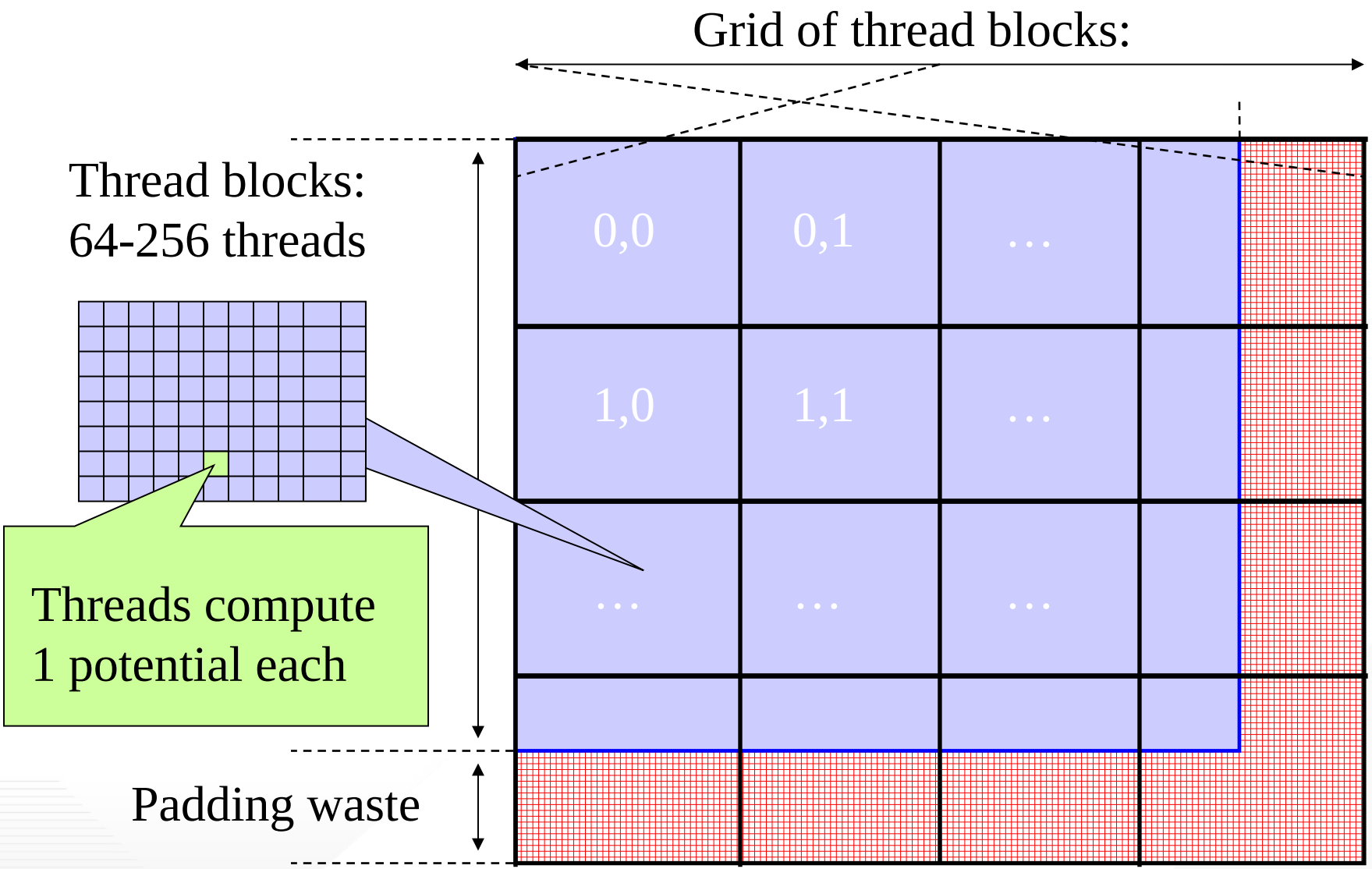
y and z distances for an atom re-calculated for all grid points More redundant work

Pros and Cons of the Slower Sequential Code

- Pros
  - Fewer accesses to the energygrid array
  - Simpler code structure

- Cons
  - More calculations on the coordinates
  - More accesses to the atom array
  - Overall, much slower sequential execution due to the larger number of calculations performed

DCS CUDA Block/Grid Decomposition (no thread coarsening)



A Fast DCS CUDA Gather Kernel

```
void __global__ cenergy(float *energygrid, dim3 grid, float gridspacing, float z, float *atoms,
   int numatoms) {
```

one thread per grid point

```
  int i = blockIdx.x * blockDim.x + threadIdx.x;
  int j = blockIdx.y * blockDim.y + threadIdx.y;
  int atomarrdim = numatoms * 4;
  int k = z / gridspacing;
  float y = gridspacing * (float) j;
  float x = gridspacing * (float) i;
  float energy = 0.0f;
  for (int n=0; n<atomarrdim; n+=4) {     // calculate potential contribution of each atom
        float dx = x - atoms[n    ];
        float dy = y - atoms[n+1];
        float dz = z - atoms[n+2];
        energy += atoms[n+3] / sqrtf(dx*dx + dy*dy + dz*dz);
      }
   energygrid[grid.x*grid.y*k + grid.x*j + i] += energy;
}
```

A Fast DCS CUDA Gather Kernel

```
void __global__ cenergy(float *energygrid, dim3 grid, float gridspacing, float z, float *atoms,
   int numatoms) {


  int i = blockIdx.x * blockDim.x + threadIdx.x;
  int j = blockIdx.y * blockDim.y + threadIdx.y;
  int atomarrdim = numatoms * 4;
  int k = z / gridspacing;
  float y = gridspacing * (float) j;
  float x = gridspacing * (float) i;
  float energy = 0.0f;
  for (int n=0; n<atomarrdim; n+=4) {     // calculate potential contribution of each atom
        float dx = x - atoms[n    ];
        float dy = y - atoms[n+1];
        float dz = z - atoms[n+2];
        energy += atoms[n+3] / sqrtf(dx*dx + dy*dy + dz*dz);
     }
   energygrid[grid.x*grid.y*k + grid.x*j + i] += energy;
}
```

All threads access all atoms in same order
Consolidated writes to grid points

Additional Comments

- Further optimizations
  - dz*dz can be pre-calculated and sent in place of z
- Gather kernel is much faster than a scatter kernel
  - No serialization due to atomic operations
- Compute efficient sequential algorithm does not translate into the fast parallel algorithm
  - Gather vs. scatter is a big factor
  - But we will come back to this point later!

Even More Comments

- In modern CPUs, cache effectiveness is often more important than compute efficiency
- The input oriented (scatter) sequential code actually has very bad cache performance
  - energygrid[] is a very large array, typically 20+ times larger than atom[]
  - The input oriented sequential code sweeps through the large data structure for each atom, trashing cache.
- The fastest sequential code is actually an optimized output-oriented code

Outline of A Fast Sequential Code

```
for all z {
  for all atoms {precompute dz² }
  for all y {
     for all atoms {precompute dy² (+ dz²) }
     for all x {
        for all atoms {
         compute contribution to current x,y,z point
            using precomputed dy² and dz²
        }
       }
    }
}
```

More Thoughts on Fast Sequential Code

- Needs temporary arrays for pre-calculated dz^2 and $dy^2 + dz^2$ values

- So, why does this code have better cache behavior on CPUs?

More Thoughts on Fast Sequential Code

- Needs temporary arrays for pre-calculated dz^2 and $dy^2 + dz^2$ values

- So, why does this code have better cache behavior on CPUs?

- Answer: the atom data structure is much smaller than the grid structure in practice.
  - Sweeping the atom data structure multiple times is more efficient since each sweep will likely find the data that was brought into cache by the previous sweep.
  - Writing high performance sequential code or parallel code is an engineering design effort, the tradeoffs often depends on data sizes

References

- Wen-mei W. Hwu, David B. Kirk, and Izzat El Hajj. *Programming Massively Parallel Processors: A Hands-on Approach*. Morgan Kaufmann, 2022.